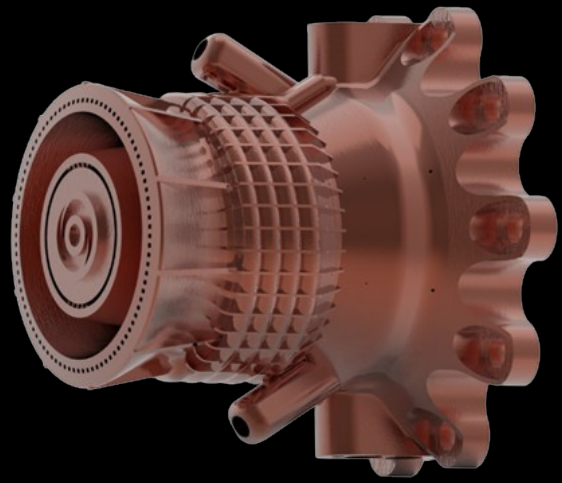
Project Wavefront Mark I RDRE

Joshua Dalal

Co-Founder and Propulsion Engineer



Project Wavefront

November 2026

RDRE Fundamentals

- **Why an RDRE?**
  - Utilizes pressure-gain combustion, increasing thermodynamic efficiency over conventional deflagration engines
  - Pressure gain enables higher chamber pressure without additional work, increasing specific impulse
  - Higher propulsion efficiency directly translates to greater payload capacity or mission range
  - Recent advances in additive manufacturing and high-temperature alloys have made practical RDREs feasible

- **Operating Principle**
  1. Fuel and oxidizer are continuously injected into an annular combustor
  2. Ignition establishes one or more rotating detonation waves
  3. Waves consume fresh propellant at **supersonic combustion speeds (>1000 m/s)**
  4. High-pressure combustion products expand through the nozzle to produce thrust

Continuous Injection → Detonation Wave → Nozzle Expansion → Thrust



Animation courtesy of Oak Ridge National Laboratory

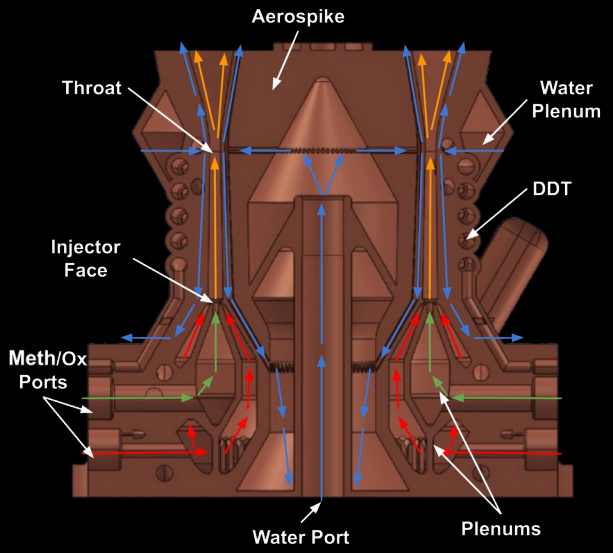
Mark I RDRE

15-person undergraduate team | Mark I development: April 2025 – Present | Inconel heat-sink article hot fired ~100 times in Nov 2026; GRCop-42 version in development

Design Objectives

- Design a 2 kN methane/oxygen rotating detonation rocket engine
- Develop a complete pressure-fed propulsion system from concept to manufacturing
- Demonstrate stable rotating detonation through hot-fire testing
- Generate experimental data to inform future RDRE development

Parameter	Value
Propellants	GOX/GCH_4
Thrust	2 kN
$P_{chamber}$	~21 bar
Predicted Isp	255 s
O/F Ratio	3.0
Flow Rate	0.8 kg/s
Material	Inconel → GRCop-42
Feed	Pressure-fed
Heat Load	0.97 MW
Annulus Diameter	2.56″



What I Owned

Mark I is a 15-person effort. These four areas are the ones I carried from the first sizing calculation through to hardware and the test plan.

Design Point

Closed the 2 kN design point using NASA's RDRE blowdown model backed by first-principles hand calculations. Iterated annulus geometry, throat and nozzle ratios and injection pressure to hit thrust and Isp while keeping chamber pressure and heat load inside feed, thermal and structural limits.

2.01 kN · 21.1 bar · Isp 255 s · 969 kW

Feed System

Sized every sonic orifice and injector from compressible-flow relations, then validated the whole path in Fluent: discharge coefficients, choking, plenum-to-injector distribution and line losses. Defined the diamond plenum geometry and determined pressure drop through regulators and valves.

Cd 0.81 → 0.96 · maldistribution within 2%

Cooling

Built and ran the conjugate heat transfer wall model for the actively cooled GRCop-42 engine, predicting local wall temperatures around the chamber and producing the temperature fields the structural analysis was run against.

575.4 K peak wall temperature

Test Preparation

Designed the nitrogen cold-flow test that verifies the feed system before ignition, sized the regulator and valves, laid out instrumentation, and coordinated with the AFRL test site on hosting the hot-fire campaign.

~100 hot fires · 10 s longest burn

System Architecture Decisions

Decision	Selected	Why	Tradeoff accepted
Feed cycle	Pressure-fed	Avoids turbomachinery, uses commercial high-pressure gas storage	Lower achievable chamber pressure and poor storage density
Propellant	GOX/GCH_4	Avoids cryogenic handling; clean and repeatable	Lower density, less flight-like
Material	Inconel 718, GRCop-42	Mature LPBF process and high-temperature strength	Short firing duration; no steady-state thermal validation
Thermal Strategy	Staged	Inconel engine validates combustion; GRCop-42 adds active cooling	Two hardware builds; added pump and plumbing
Chamber Pressure	~21 bar	Selected feasible point within feed, thermal, and structural limits	Lower performance than a higher-pressure design

The architecture prioritized testability, manufacturability, and rapid learning over maximum flight performance.

Combustion Chamber Sizing

I used NASA's RDRE Blowdown Model, supported by first-principles hand calculations, to size the engine and close the design point. I iterated key geometry and operating parameters to achieve the target thrust and specific impulse while keeping chamber pressure and thermal loads within feasible system limits.

Inputs	
Mixture Ratio	3
Injection Pressure	8.411 bar
Throat Ratio	0.73
Nozzle Ratio	3.6
Engine Diameter	2.56 in
Inner/outer diameter ratio	0.87



Outputs	
Frequency	13.6 kHz
Thrust	2.01 kN
Isp	255 s
Mass Flow	0.816 kg/s
Chamber Pressure	21.1 bar
Chamber Temp	4102 K
Heat Load	968.8 kW
Wall Heat Flux	52.5 MW/m^2
Chamber Length	1.78 in
Chamber Inner Diameter	2.38 in
Chamber Outer Diameter	2.74 in

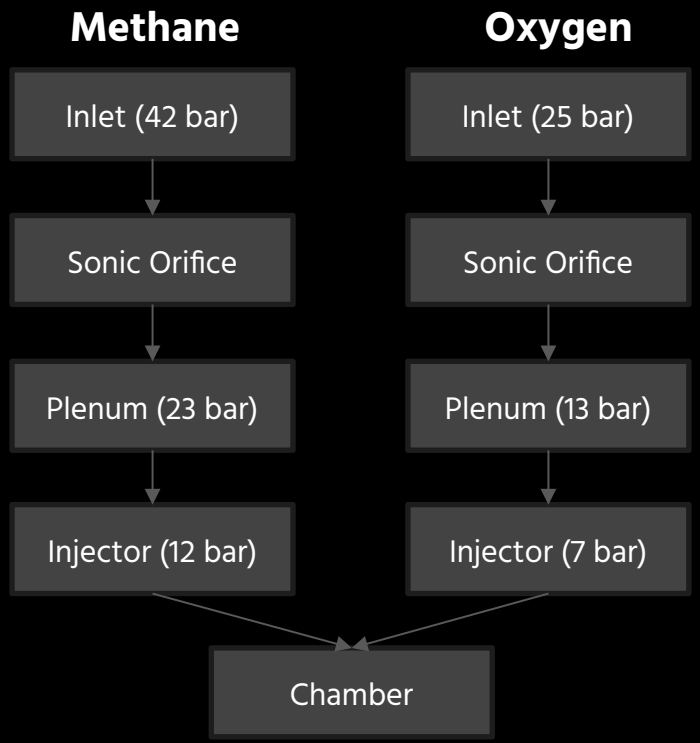
Feed System Architecture

System Goals and Limitations

- Deliver required methane and oxygen mass flow at desired pressures
- Emphasis on reliable mass flow at varying chamber pressures
- Pressures/mass flows under structural limits
- Injector and plenum geometry limited by printing capabilities

Methods and Validation Techniques

- Sonic orifices near the ports of the engine for constant mass flow independent of downstream pressure
- Choked injectors to minimize backflow and improve refill time
- Two orifices for each propellant to minimize maldistribution around the injector face
- Built custom scripts and Excel sheets
- Used CFD to validate mass flow, choking, pressure loss, and distribution



Propellant Delivery Sheet

- Assumptions:
  - Ideal gas, constant k, γ, and R
  - Linear Joule–Thomson cooling, $\Delta T = \mu_{JT} \times \Delta P$
  - Steady, adiabatic, uniform, 1-D flow, no transients
  - Negligible line friction losses (confirmed analytically)
- Logic:
  - Input regulator inlet P0 , T0, gas properties, and mass flow
  - Use Joule-Thomson to find regulator outlet temperature
  - Use choked-flow relations to compute orifice exit conditions
  - Find $\dot{m}$ per orifice, back-calculate area and diameter
  - Compute injector-exit state P3, T3, ρ3, and momentum flux
- Equations:
  - Choked flow equation: $\dot{m} = C_d \frac{A_0 P_0}{\sqrt{T_0}} k$ with $k = \sqrt{\frac{\gamma}{R}} \left(\frac{\gamma+1}{2}\right)^{-\frac{\gamma+1}{2(\gamma-1)}}$
  - Joule Thomson Effect (regulator): $T_2 = T_1 + (P_2 - P_1) * u_{JT}$
  - Pressure/temp drop through orifices: $P_2 = \frac{P_1}{(1+\frac{\gamma-1}{2})^{\frac{\gamma}{\gamma-1}}}$, $T_2 = T_1 \left(\frac{2}{\gamma+1}\right)$
- Results:
  - Orifice diameter, injector diameter, injector exit conditions

Important Results	
Methane Port Pressure	42.09 bar
Methane Orifice Diameter	4.279 mm
Methane Injector Pressure	12.453 bar
Methane Injector Diameter	0.680 mm
Oxygen Port Pressure	25.313 bar
Oxygen Orifice Diameter	7.931 mm
Oxygen Injector Pressure	7.064 bar
Oxygen Injector Diameter	1.734 mm

See appendix for full propellant delivery sheet and line friction losses

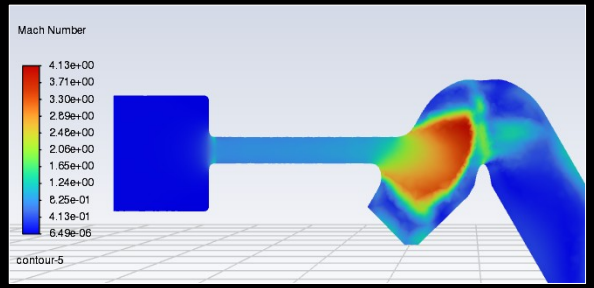
Port Orifice Design

General equation used: $\dot{m} = C_d \frac{A_0 P_0}{\sqrt{T_0}} k, \quad \text{with} \quad k = \sqrt{\frac{\gamma}{R}} \left(\frac{\gamma+1}{2}\right)^{-\frac{\gamma+1}{2(\gamma-1)}}$

- Discharge coefficient, C_d, was determined through CFD, and improved through fillets and chamfers

Steps for finding C_d

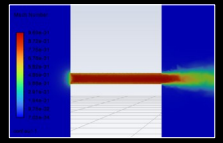
1. Tested initially with a diameter set by choosing $C_d = 1$
2. Given the mass flow from the simulation, back solve what the actual C_d value is with the choked flow equation: $C_d = \frac{\dot{m}\sqrt{T_0}}{A_0 P_0 k}$
3. Resolved for the diameter needed for the required mass flow



Orifice	C_d
Methane Port	0.9401
Oxygen Port	0.9337
Methane Ignition	0.7215
Oxygen Ignition	0.8513

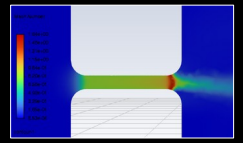
Theoretical mass flow: 0.170114kg/s

No Fillet



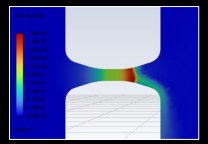
Mass flow: 0.139206kg/s
Calculated Cd: 0.81012

Fillet



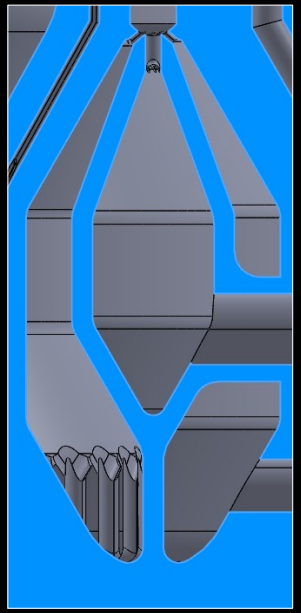
Mass flow: 0.166183kg/s
Calculated Cd: 0.96213

Chamfer + Fillet



Mass flow: 0.169198kg/s
Calculated Cd: 0.98467

Plenum Design

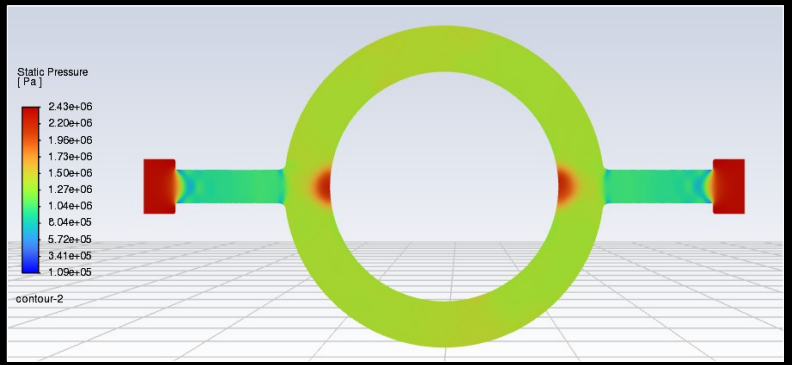


Diamond Shape

- Basic structure modeled after standards for additively manufactured RDREs
- Maximizes area to minimize maldistribution while being printable
- Width is maximum under engine diameter
- Height minimized for print costs, but big enough to fit sensor/feed ports

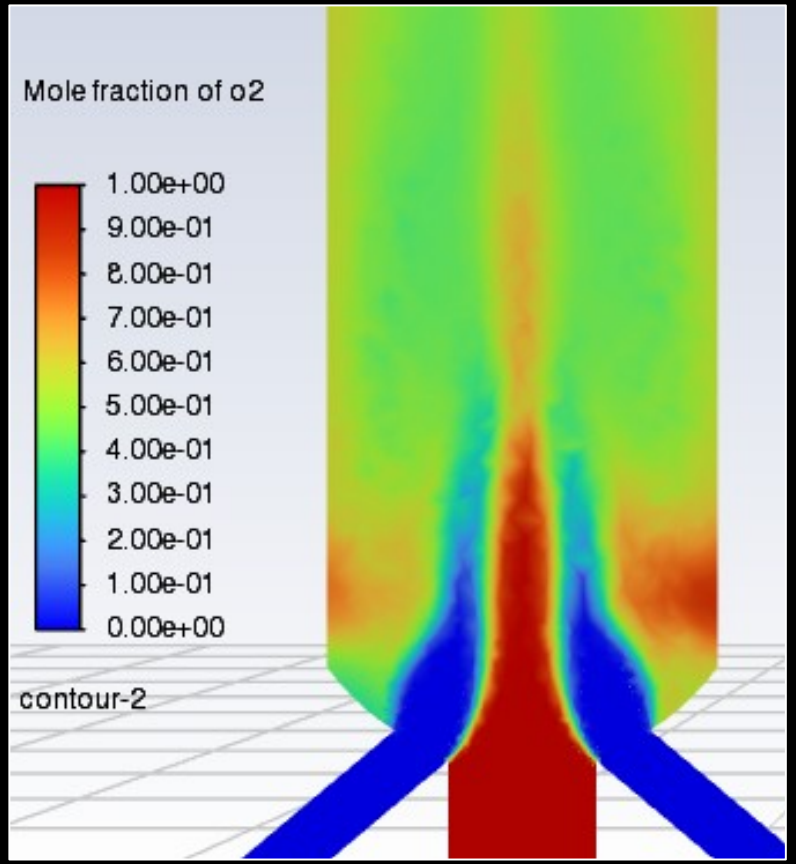
CFD Analysis

- Performed by tracking mass flow through every injector
- Oxygen:
  - Minimum: 8.86 g/s
  - Maximum: 9.04 g/s
  - % Difference: 1.99%
- Methane:
  - Minimum: 1.444 g/s
  - Maximum: 1.470 g/s
  - % Difference: 1.30%



Injectors

- Requirements
  - Manufacturing simplicity
  - Good mixing quality
  - Keep injection pressure within feed range
  - Axial momentum flux
- Design
  - 70 choked triplet impinging injectors, F-O-F
  - 50-degree impingement angle
    - Maximizes mixing while keeping momentum flux axial
  - Methane injectors
    - 0.680 mm diameter, 12.453 bar
    - C_d = 0.9102
  - Oxygen injectors
    - 1.734 mm diameter, 7.064 bar
    - C_d = 0.9646
- Validation Momentum flux comparison
  - Ox: 1.187 MPa
  - Methane: 1.211 MPa
  - Momentum flux difference: 2.01%



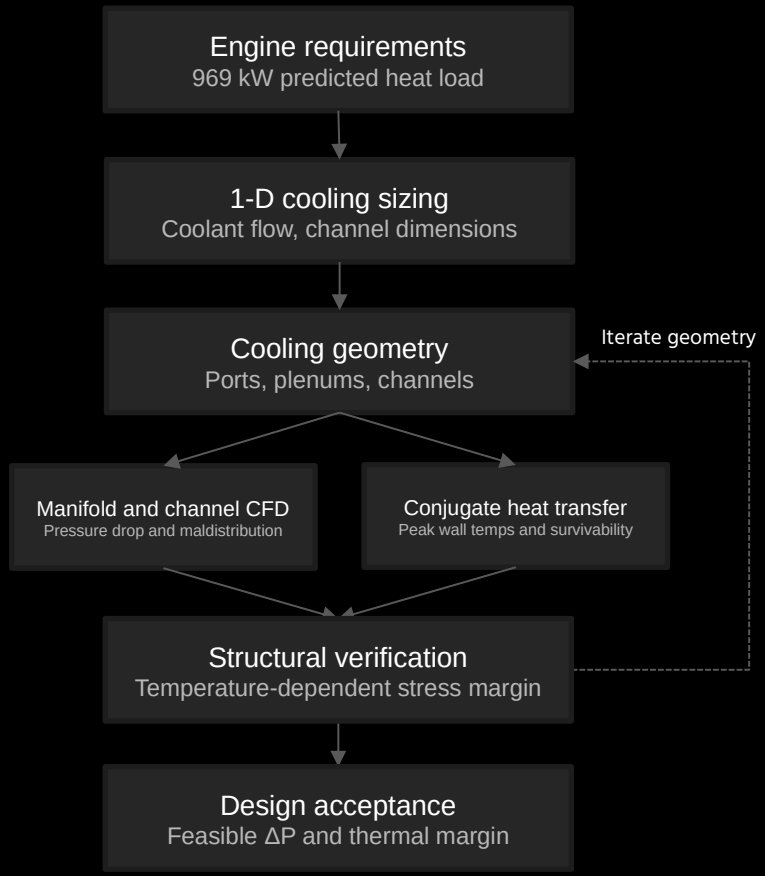
Active-Cooling Development

Requirements

- Reject approximately 969 kW of predicted chamber heat load
- Maintain wall temperatures within material and structural limits
- Minimize coolant pressure drop to enable a practical pump system
- Maintain acceptable flow distribution across all cooling paths
- Fit within the engine envelope and LPBF manufacturing constraints
- Provide temperature fields for structural analysis

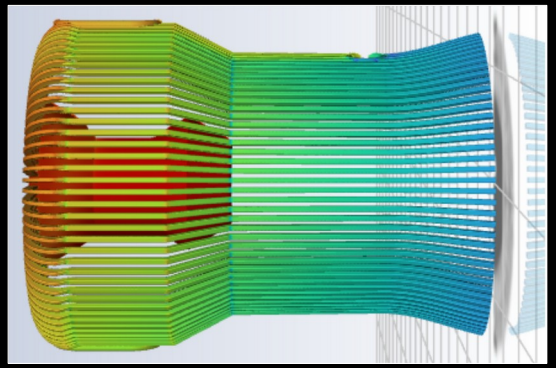
Analysis Methods

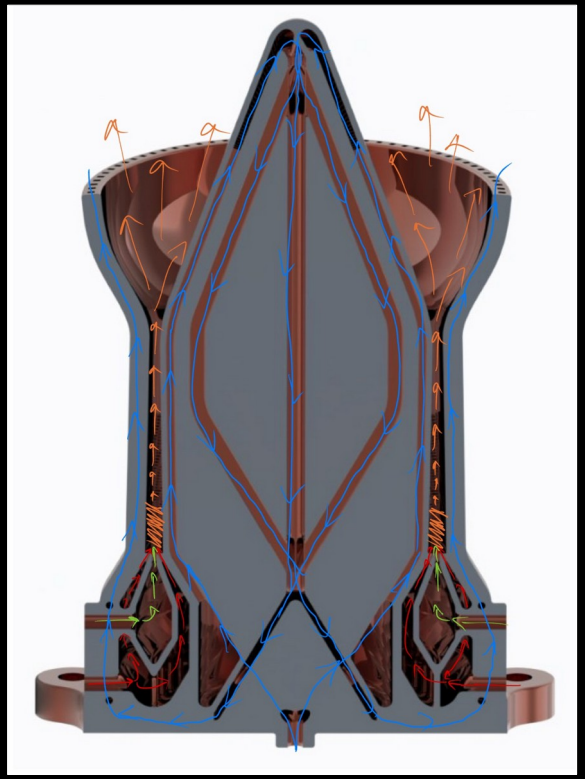
- Analytical model sized channel flow and estimated wall temperature
- CFD evaluated manifold pressure drop and coolant maldistribution
- Conjugate heat transfer predicted local wall temperatures
- Temperature distributions were sent to structural teams



Cooling Architecture Trades

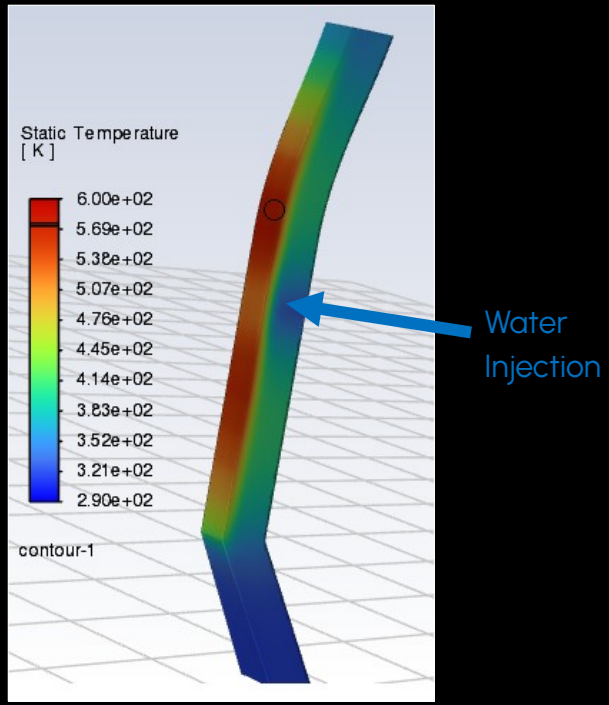
- One port at the base of the engine
  - Water loops through the top of the engine
  - Just one flow path
  - Simple design, but had insurmountable pressure drop (~130 psi)
- Second iteration split the water at the base
  - One path up the core of the engine and one around the chamber shroud
  - Reduced pressure drop but still high (~85 psi)



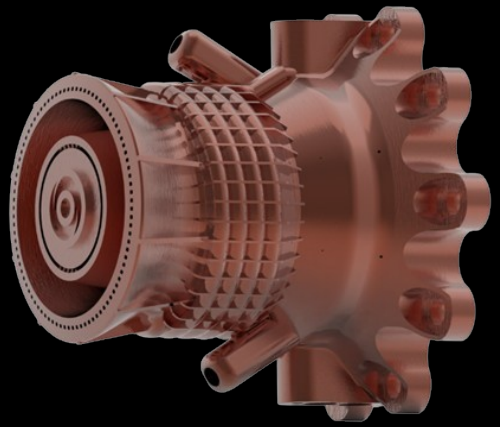


Selected GRCop-42 Cooling Geometry

- Water is injected at ¾ chamber length
- 0.04 kg/s per cooling channel
- Results from this simulation were provided to FEA team for structural analysis
- Benefits
  - Peak wall temperature of 575.4 K preserves greater material strength
  - Pressure drop of 42 psi is manageable
- Drawbacks
  - Smaller cooling channels are more sensitive to manufacturing constraints
  - Requires a secondary injection plenum around the engine



From CAD to Hardware



CAD: single-piece LPBF



Printed Inconel 718 article

Heat-sink article: 6.8 in long, 6 in OD, 7.2 kg.
10x Ø0.375 mounting holes at 36° spacing, 4x 1/4-18 NPT feed ports.

LPBF constraint → design response

Sharp-edged orifices lose flow

Chamfered and filleted port entries; Cd 0.81 → 0.96

Overhang and self-support limits

Diamond plenum cross-section, printable without supports

Post-processing access is limited

1/4-18 NPT ports left unmachined and adapted at the stand

Single-piece build

All feed passages internal; no sealed joints in the high-pressure path

Predicted Performance

The shared combustion and flow architecture is closed. The Inconel heat-sink article has since been hot fired (slides 20 to 23); the table shows pre-test predictions, and the GRCop-42 active-cooling step remains in development.

Shared Mark I Architecture		
Predicted thrust	2.01 kN	Performance model
Total propellant flow	0.816 kg/s	Analytical sizing
Average chamber pressure	21.1 bar	Performance model
[redacted]	~0.96	CFD
Propellant maldistribution	GOX 1.99%, GCH_4 1.30%	CFD
Inconel Heat-Sink Step		
Engine article	Manufactured	Physical hardware
Integrated hot fire	Pending	Experimental validation
GRCop-42 Active-Cooling Step		
Peak wall temperature	575.4 K	CHT
Cooling pressure drop	42 psi	CFD
Cooling manifold	In development	Design status



Validation Status Before Hot Fire

Hand calcs	Analytical sizing	CFD	Printed article	Cold flow	Hot fire	Correlation
complete	complete	complete	complete	in preparation	AFRL, Nov 4	planned

Closed in Analysis

Design point closes at 2.01 kN, 0.816 kg/s, 21.1 bar against the blowdown model and hand calculations

Predicted Cd raised from 0.81 to 0.96 through CFD-driven geometry changes

Plenum distribution within 2.0% (GOX) and 1.3% (GCH_4) across 70 injector elements

Injector momentum flux matched to within 2.0%

The Inconel heat-sink article prints as designed

What is not proven yet

No combustion data. Detonation stability, wave frequency and real wall heat flux are all predictions

Every Cd is CFD-derived, not flow-tested on printed hardware

Cooling numbers (42 psi, 575.4 K) are CFD and CHT results on a part that does not exist yet

Ignition and startup reliability are untested

Outcome: the Mark I completed ~100 hot fires in November 2026, the longest lasting 10 seconds, with detonation in 2, 3 and 4-wave modes. Results follow on slides 20 to 23.

Cold-Flow Test Development

Nitrogen path

Same passages, same ports, nitrogen instead of propellant.

- **N_2 supply**
  high-pressure bottle, blowdown
- **Two-stage regulator**
  sizing now
- **Valve**
  Cv and response, sizing now
- **Sonic orifices**
  Ø4.28 / Ø7.93 mm, as printed
- **Plenum**
  diamond, side-to-side split
- **Injectors**
  70 × F-O-F triplet, to ambient

Measurement chain

1. Measure nitrogen mass flow against inlet pressure across a sweep, not a single point.
2. Back out effective CdA at each port and injector from the choked-flow relation.
3. Compare with the values the model runs on.
4. Rescale to oxygen and methane, then correct the delivery sheet before the engine is fired.

Open engineering right now

Regulator and valve Cv — sizing for the required flow at 25 to 42 bar port pressure.

Instrumentation — transducer ranges and tap locations at supply, orifice inlet and plenum.

Test hardware — nitrogen adapters for the 4x 1/4-18 NPT ports, interchangeable configuration.

AFRL hot fire, November 4 nitrogen cold flow first to measure the feed system, then straight into the hot-fire campaign.

Hot-Fire Test Plan

November 4 · Inconel heat-sink article · short-duration firings on the AFRL stand

Series	What varies	Set points	Firing	What it settles
01 Ignition	Nothing. The design point, repeated.	O/F 3.0 (φ 1.33), 0.816 kg/s, Pc ≈21 bar	0.1 s	Whether the DDT system lights it and a wave forms and rotates.
02 Flow sweep	Feed pressure, which sets mass flow. Chamber pressure follows.	≈0.6 to 0.9 kg/s, Pc ≈16 to 23 bar, O/F held at 3.0	0.1 to 0.3 s	Chamber pressure against the sizing sheet; c* and wave frequency versus Pc.
03 Equivalence ratio	Oxidizer-to-fuel split at fixed total flow.	φ 1.0 to 1.6 (O/F 4.0 to 2.5)	0.1 to 0.3 s	Detonability limits, wave count, and where performance peaks.

What we measure it with

CTAP pressure
Capillary Tube Averaged Pressure. The tube damps the oscillation so a recessed transducer reads stable chamber and plenum pressure, but it will not resolve the wave.

250 kfps camera
Wave count and rotation rate. Roughly 18 frames per cycle at the predicted 13.6 kHz.

Load cells
Thrust, and Isp and c* once combined with measured mass flow.

Thermocouples
Wall temperature rise and heat soak against the heat-sink model.

Hot-Fire Results

November 2026 · Mark I · gaseous oxygen with methane and ethylene

~100
Hot fires

10 s
Longest burn

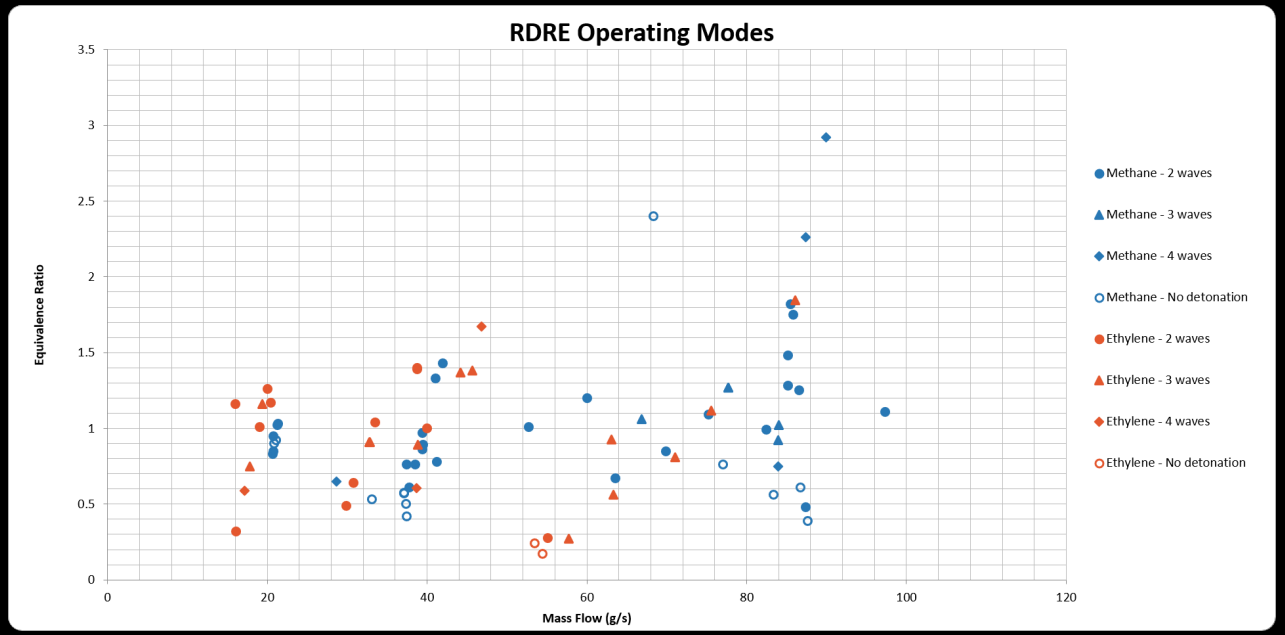
2 to 4
Detonation waves observed

2
Fuels tested: methane, ethylene



To our knowledge, the longest-duration RDRE hot fire by a student group.

Operating Map



Wave count by mass flow and equivalence ratio, every November 2026 test

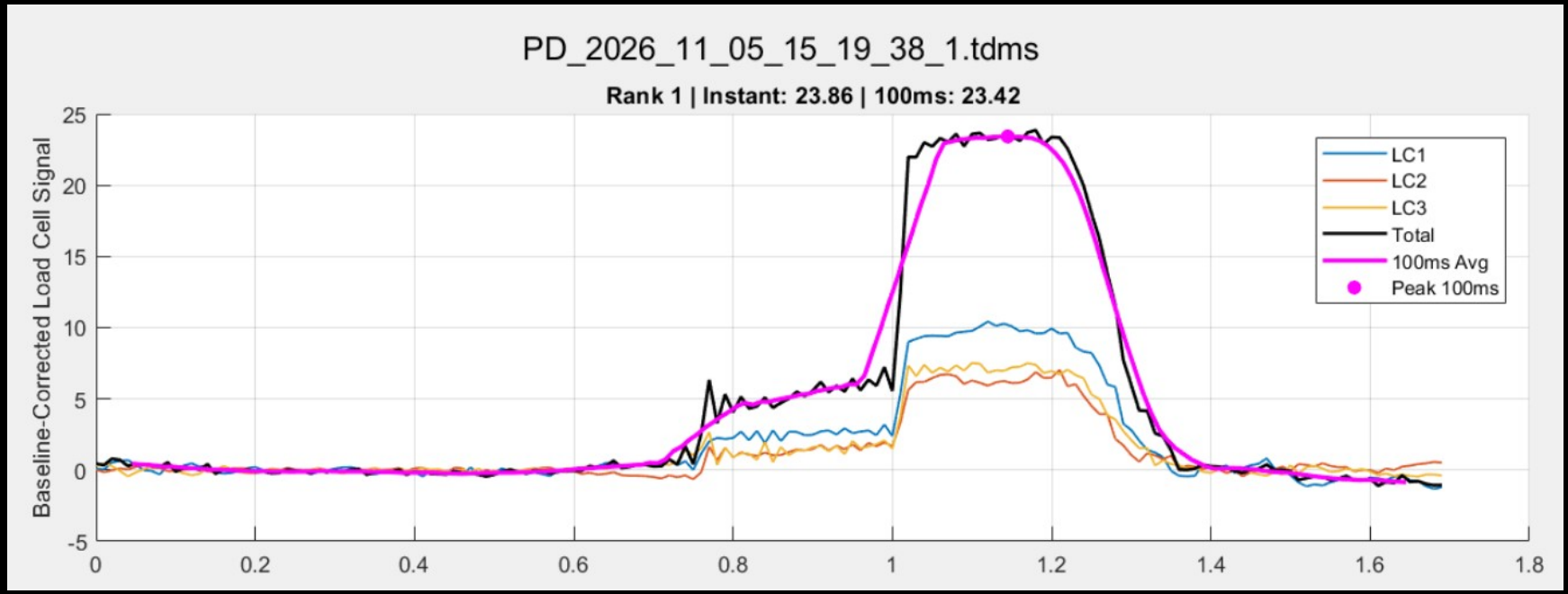
What the map shows

Methane and ethylene both sustained 2, 3 and 4-wave detonation.

Tests spanned roughly 16 to 97 g/s, well below the 0.816 kg/s design flow.

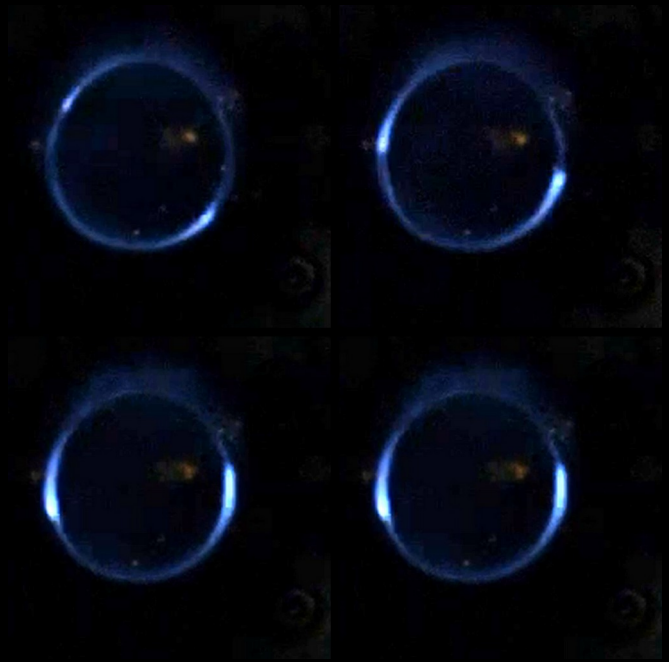
Most methane runs that failed to detonate were lean, below φ 0.8.

Thrust Measurement



Baseline-corrected signal from three load cells and their total, Nov 5 test, with the 100 ms moving average and its peak

Inside the Annulus



Successive high-speed frames looking up the annulus: two detonation fronts (brightened)



Injector orifice ring, backlit through the annulus



Aerospike after hot fire

Lessons Learned

Four things I would do differently, and what each one cost us.

Close constraints first

We targeted a larger engine before closing feed pressure, gas inventory, cooling capacity and safety constraints, then had to walk the design point back.
Next time: build the facility capability envelope first, then pick the best design point inside it.

Manufacturing limits are requirements

Minimum radii, port spacing, overhang angles and post-processing access forced repeated geometry changes after CAD was largely done.
Next time: an LPBF design-rule checklist and representative print coupons before detailed design.

Validate sensitive assumptions early

Assuming Cd = 1 overpredicted port flow by roughly 22%. CFD-driven fillets and chamfers recovered it, but only after the geometry was set.
Next time: flow-test printed orifices and channels before committing to the full engine print.

Optimize the system, not the part

The simplest cooling geometry gave an impractical 130 psi drop. Added manifold complexity brought it to 42 psi at an acceptable wall temperature.
Next time: couple thermal, pressure-drop, structural and manufacturing models earlier, with explicit system-level acceptance criteria.

Thank you



Appendix

Design drivers

Design methodology

Future work and Mark II

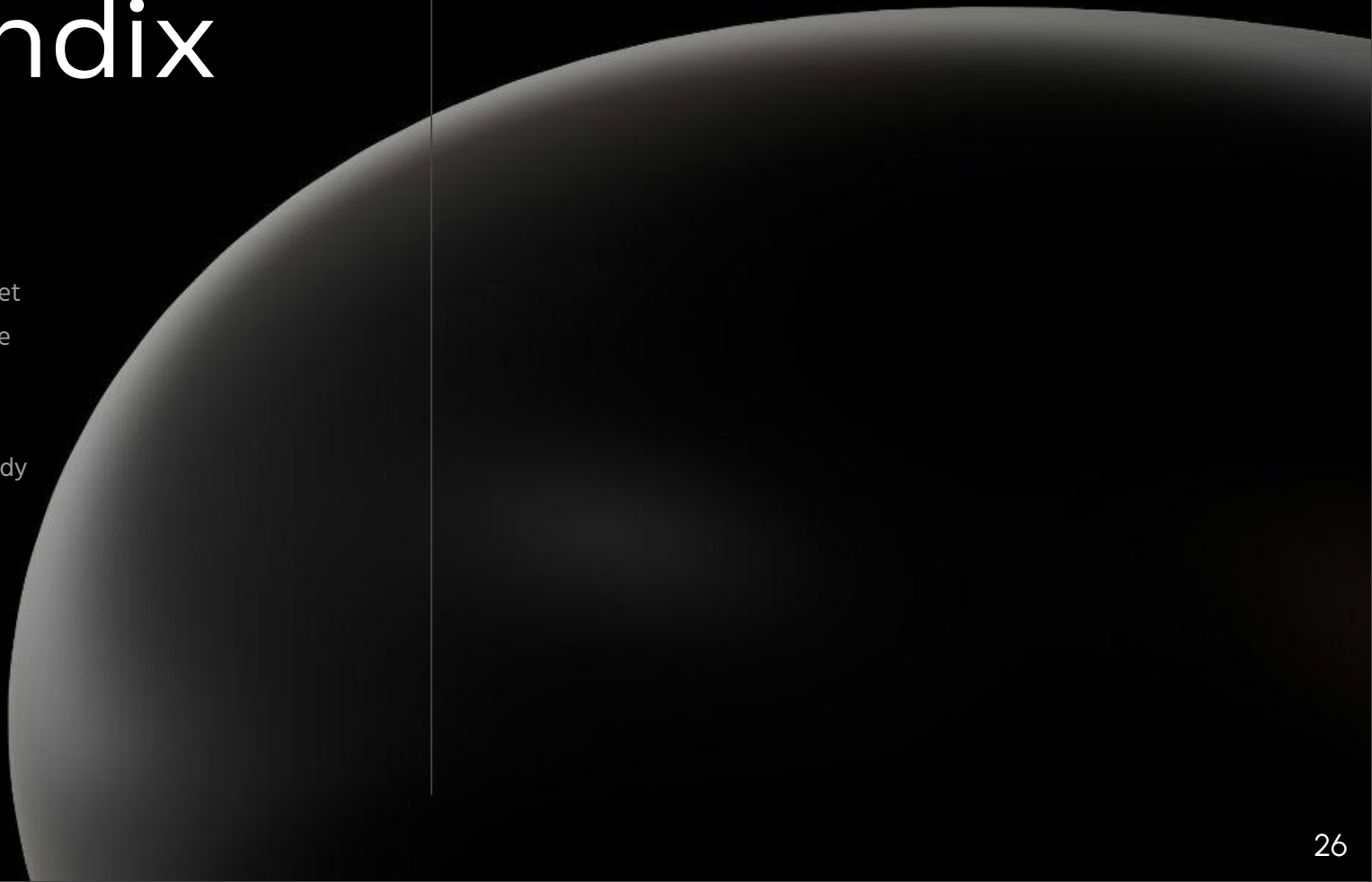
Full propellant delivery sheet

CFD setup and convergence

Line friction losses

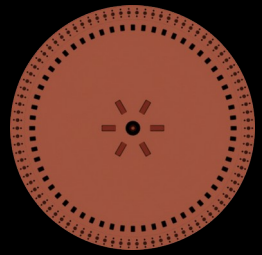
DDT ignition flow system

Cooling maldistribution study

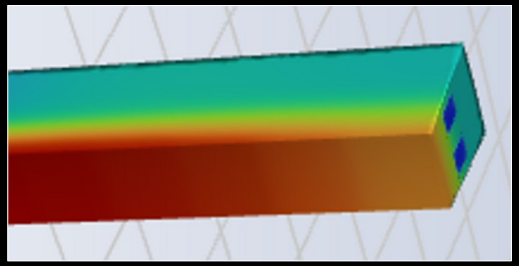


Design Drivers

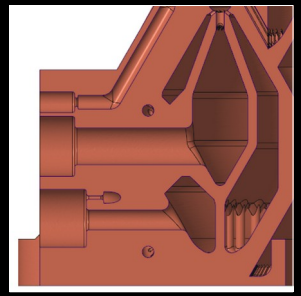
Flow Control
Precise methane and oxygen mass flow were required to sustain stable rotating detonation.



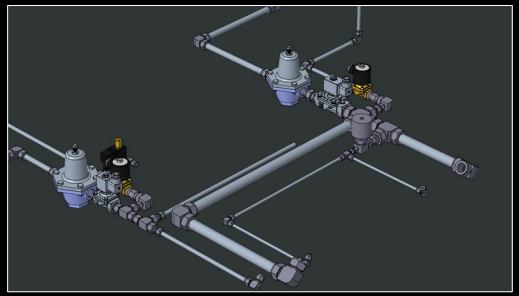
Thermal Management
The chamber had to survive nearly 1 MW of heat load while remaining structurally sound.



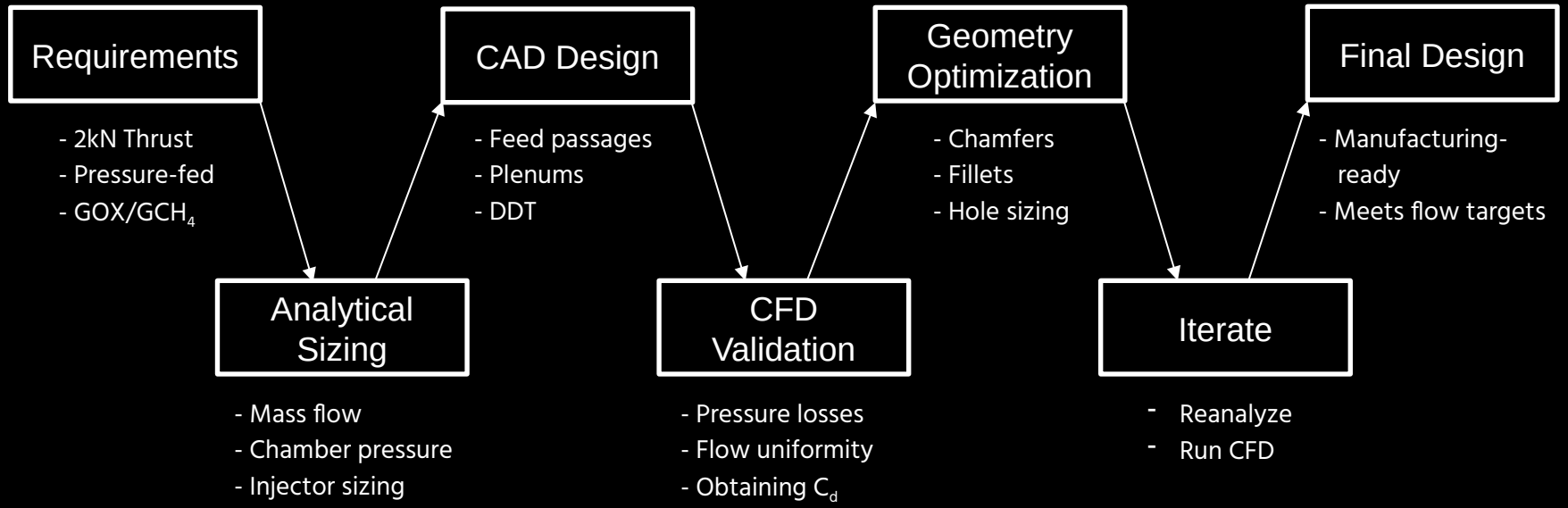
Manufacturability
Internal passages had to be printable as a single-piece metal component with minimal post-processing.



Testability
Designed for straightforward instrumentation and experimental validation.



Design Methodology



Future Work

Testing

The initial hot-fire campaign is complete: ~100 tests in November 2026, burns up to 10 seconds, and detonation in 2, 3 and 4-wave modes. Next is correlating that data against the sizing tools.

Model Validation

Compare experimental mass flow rates and chamber pressures with analytical and CFD predictions. Use the results to refine the sizing tools and improve future design accuracy.

Performance Characterization

Measure thrust, specific impulse, chamber pressure, and detonation frequency to quantify engine performance. Correlate these measurements with thermal behavior to validate the overall design.

Mark II Improvements

After validating the Mark I GRCop-42 engine, develop a higher-performance regeneratively cooled successor with increased chamber pressure and longer-duration capability.

Full Propellant Delivery Sheet

Changing Diameter

Mass Flow Requirements

Gas Constants

$$k = \sqrt{\frac{\gamma}{R}}\left(\frac{\gamma+1}{2}\right)^{-\frac{\gamma+1}{2(\gamma-1)}}$$

Choked flow equation: $P^* = \frac{\dot{m}\sqrt{T_0}}{A_0 k C_d}$

INJECTION PRESSURE

Calculates temperature drop through regulator (based on pressure estimate)

$$T_2 = T_1 - (P_2 - P_1) * u_{JT}$$

Recalculates required pressure

$$P_2 = \frac{P_1}{(1+\frac{\gamma-1}{2})^{\frac{\gamma}{\gamma-1}}}$$

$$T_2 = T_1\left(\frac{2}{\gamma+1}\right), \quad a = \sqrt{\gamma R T}$$

Choked flow equation: $A = \frac{\dot{m}\sqrt{T_0}}{P k C_d}$

$$P_2 = \frac{P_1}{(1+\frac{\gamma-1}{2})^{\frac{\gamma}{\gamma-1}}}, T_2 = T_1\left(\frac{2}{\gamma+1}\right)$$

Propellant Feed System

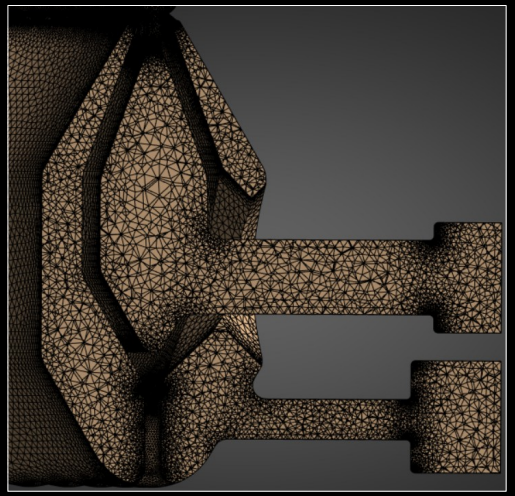
Fuel		Fuel (Ignition)		Oxidizer		Oxidizer (Ignition)	
d	4.27892 mm	P1 (from fuel)	4209105.856 Pa	d	7.931375 mm	P1 (from ox)	2531275.909 Pa
A	1.438E-05 m^2	T1 (from fuel)	268.5278189 K	A	4.94068E-05 m^2	T1 (from ox)	262.5141458 K
mdot fuel	0.2041164 kg/s	mdot fuel	0.000193911 kg/s	m dot ox	0.6123492 kg/s	mdot ox	0.000918524
mdot fuel (half)	0.1020582 kg/s	mdot fuel (one DDT)	9.69553E-05 kg/s	m dot ox (half)	0.3061746 kg/s	mdot ox (one DDT)	0.000459262 kg/s
T0	293 K	T0	293 K	T0	293 K	T	293 K
γ	1.31	γ	1.31	γ	1.4	γ	1.4
R	518.2 J/kgK	R	518.2 J/kgK	R	259.8 J/kgK	R	259.8 J/kgK
M	16.04 gmol	M	16.04 gmol	M	32 gmol	M	32 gmol
Choked Flow Factor (k)	0.029391286 Units	Choked Flow Factor (k)	0.029391286 Units	Choked Flow Factor (k)	0.042481584 Units	Choked Flow Factor (k)	0.042481584 Units
Discharge Coefficient (Cd)	0.940104	Discharge Coefficient (Cd)	0.8564	Discharge Coefficient (Cd)	0.933723895	Discharge Coefficient (Cd)	0.851305889
P*	4396722.097 Pa	A	1.50E-08 m^2	P*	2674218.818 Pa	A	8.13E-08 m^2
P*	43.96722097 bar	d	0.138180128 mm	P*	26.74218818 bar	d	0.321707015 mm
	12.45291841 bar	P3	22.89446479 bar	P3	7.064326502 bar	P3	13.37226962 bar
Assumes Constant T^^^^. Correct for that by estimating the temperature drop from the regulator							
P0	200 bar	P0	200 bar	P0	250 bar	P0	250 bar
P1	43.96722097 bar	P1	43.96722097 bar	P1	26.74218818 bar	P1	26.74218818 bar
T0	293 K	T0	293 K	T0	293 K	T0	293 K
Joule Thomason Coef. (uJT)	0.15684 K/bar	Joule Thomason Coef. (uJT)	0.15684 K/bar	Joule Thomason Coef. (uJT)	0.13655 K/bar	Joule Thomason Coef. (uJT)	0.13655 K/bar
T1	268.5278189 K	T1	268.5278189 K	T1	262.5141458 K	T1	262.5141458 K
rho1	30.24845207 kg/m^3	rho	30.24845207 kg/m^3	rho	37.11484305 kg/m^3	rho	37.11484305 kg/m^3
P1	4209105.856 Pa	Using the same equation:	4209105.856 Pa	Using the same equation:	2531275.909 Pa	Using the same equation:	2531275.909 Pa
P1	42.09105856 bar	Upstream pressure:	42.09105856 bar	Upstream pressure:	25.31275909 bar	Upstream pressure:	25.31275909 bar
Conditions at the exit of the first sonic orifice							
P2	2289446.479 Pa	P2	2289446.479 Pa	P2	1337226.962 Pa	P2	1337226.962 Pa
P2	22.89446479 bar	P2	22.89446479 bar	P2	13.37226962 bar	P2	13.37226962 bar
T2	232.4916181 K	T2	232.4916181 K	T2	218.7617882 K	T2	218.7617882 K
rho2	19.00315832 kg/m^3	rho2	19.00315832 kg/m^3	rho2	23.52851476 kg/m^3	rho2	23.52851476 kg/m^3
v or a	397.2720416 m/s	v or a	397.2720416 m/s	v or a	282.07807 m/s	v or a	282.07807 m/s
Injector geometry calculations							
Number of injectors	140			Number of injectors	70		
Angle between injectors	5.142857143 degrees			Angle between injectors	5.142857143 degrees		
Distance between injectors	0.114892531 mm			Distance between injectors	0.114892531 mm		
Mass flow per injector	0.001457974 kg/s			Mass flow per injector	0.008747846 kg/s		
Desired velocity	1 mach			Desired velocity	1 mach		
Desired velocity	397.2720416 m/s			Desired velocity	282.07807 m/s		
Discharge Coefficient	0.910161491			Discharge Coefficient	0.964614555		
Area	3.62983E-07 m^2			Area	2.36117E-06 m^2		
Diameter	0.000679826 m			Diameter	0.001733879 m		
Diameter	0.679826437 mm			Diameter	1.733878503 mm		
Injector exit conditions based on choked flow							
T3	201.2914443 K			T3	182.3014901 K		
P3	1245291.841 Pa			P3	706432.6502 Pa		
P3	12.45291841 bar			P3	7.064326502 bar		
rho3	11.93846301 kg/m^3			rho3	14.91562301 kg/m^3		
Momentum Flux	1211133.228 kg/ms^2			Momentum Flux	1186806.852 kg/ms^2		

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CFD Setup and Checks

CFD use cases

- Effective discharge coefficient of nonideal passages
- Total mass flow at design conditions
- Plenum-to-injector flow distribution
- Local pressure losses and separation
- Cooling-system pressure drop and maldistribution



Numerical check	Acceptance criterion
Mass imbalance	
Residual convergence	
Outlet-flow stability	No meaningful drift
Mesh refinement	
Choking confirmation	

Simulation Setup

- ANSYS Fluent
- Compressible ideal gas
- Energy equation enabled
- SST k-ωturbulence model
- Pressure-based coupled solver
- Methane and oxygen modeled independently
- Inlet stagnation pressure and temperature
- Outlet pressure boundary conditions

Friction Losses

Pressure drop through ports

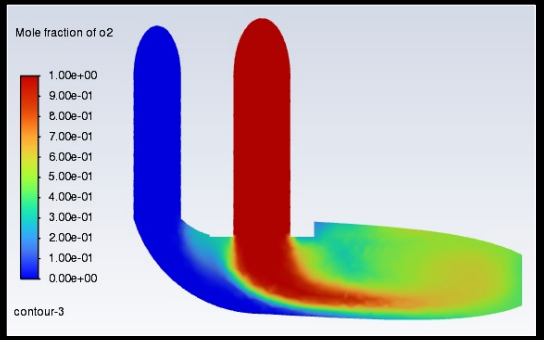
- Goal: Calculate pressure drop in the fluid as it moves through the ports into the plenums
- Assumptions: Ideal gas, fully turbulent internal flow, adiabatic, density and velocity approximately constant along the tube
- Inputs: Fluid properties, tube geometry, inlet conditions, surface roughness
- Equations used
  - Darcy Weisbach $\Delta P = f \frac{L}{D_h} \frac{\rho V^2}{2}$
  - Relative roughness $\varepsilon_r = \frac{\varepsilon}{D_h}$
  - Swamee-Jain Equation $f = \frac{0.25}{[\log(\frac{\epsilon}{3.7d} + \frac{5.74}{Re^{0.9}})]^2}$
- Outputs:
  - Darcy friction factor, frictional pressure drop, outlet pressure

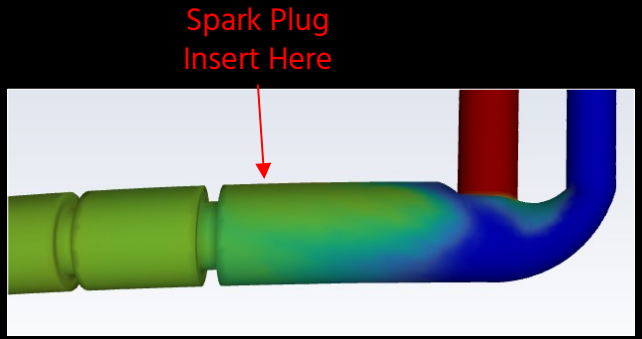
Location	Pressure Drop
Main O_2 Line	0.049 bar
Main CH_4 line	0.094 bar
O_2 Ignition	0.005 bar
CH_4 Ignition	0.011 bar

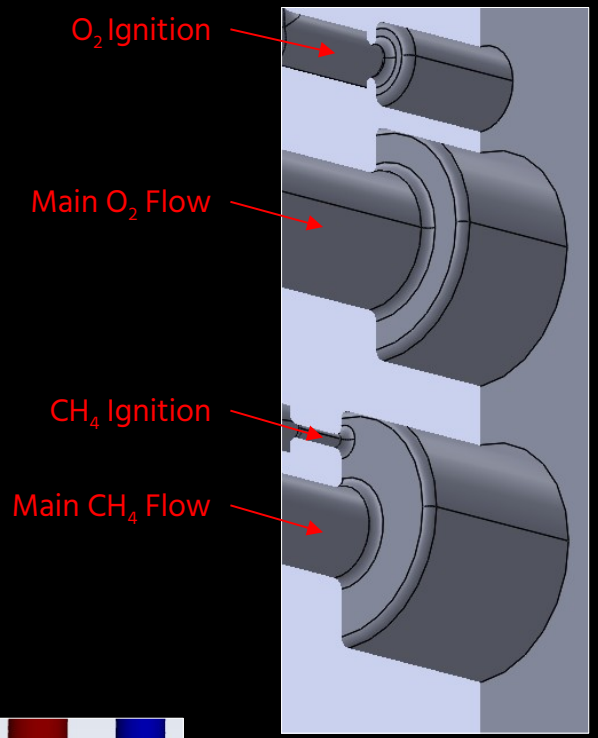
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DDT Flow System

- Methane feeds from same port as main fuel delivery
  - Stays on at all times
  - Costs slight efficiency, but ignition mass flow is <0.1% of combustion mass flow
  - Saves costs on valves
- Ox feeds from same line as main ox delivery
  - Branches off before the engine with a solenoid valve
  - Enables us to shut off ox after ignition to prevent heating in DDT
- Mixing is promoted through perpendicular injection upstream of the spark plug







Cooling Maldistribution Study

- Goal: design a plenum to distribute cooling water to all channels
  - Optimized first for maldistribution, then for pressure drop
- Required range determined through CHT → FEA pipeline
  - Nominal wall temp is 575.4 K
  - For 10 % additional flow, temp drops to 562.2 K
  - For 10 % less flow, temp rises to 586.6 K
  - Team structural analyses using these bounding temperatures remained within the selected acceptance margin
- Cooling-manifold optimization remains ongoing

Nominal Flow	40.0 g/s
Maximum	43.1 g/s
Minimum	36.1 g/s
Standard Deviation	1.86 g/s
Coefficient of Variation	4.61 %

